SEMINAR REPORT · 7 MINUTES

Reverse typography: three seminars

OCR in the age of VLMs, the printer behind a book, segmentation we can explain — read through a programme of our collaborators.



Mathis Wauquiez

Lingnan University

WHAT I FOLLOWED

Three seminars, one thread

SEMINAR · 10 OCT 2025

What is OCR really worth today?

De la vision par ordinateur aux LLMs : que vaut vraiment l'OCR aujourd'hui ?

ILB Data Lab × Centre Borelli × IRISA, one hour online.
S. Mowlavi, M. Gardella (Centre Borelli)

GTTI SEMINAR · 26 MAR 2026

Analyzing historical documents beyond text transcription

D. Belzarena.
Comedias sueltas: Spanish Golden-Age plays printed as cheap pamphlets. Who printed them?

DEFENSE · 29 SEP 2026

Mathematical formalization of image segmentation

R. W. Leon (PhD, supervised by J.-M. Morel and Z. Liu).
Towards a global abstract segmentation model: energies for SAM-like models, trained on images without semantics.

Collaborators: Seginus Mowlavi and Marina Gardella (Centre Borelli), and Diego Belzarena. For each talk: what it does, one strength, one open question.

Reverse typography: run the press backwards

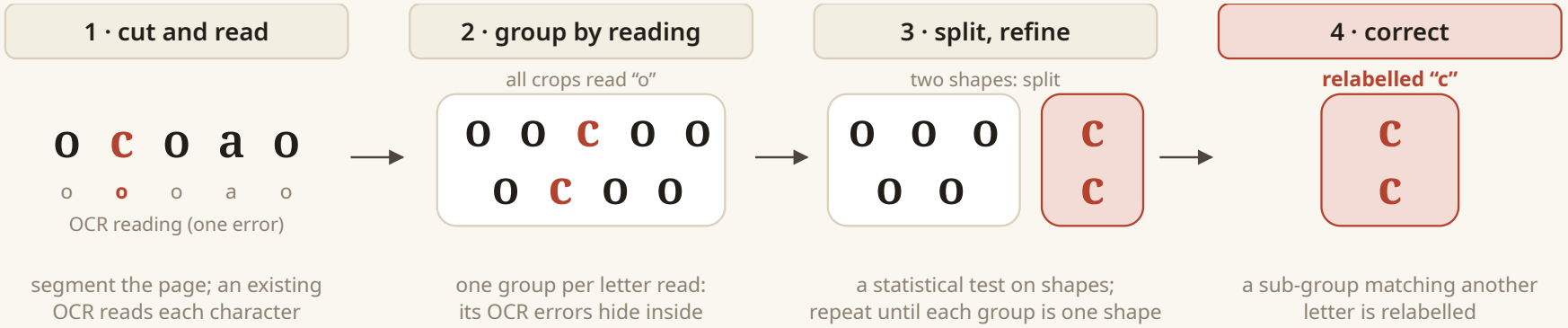


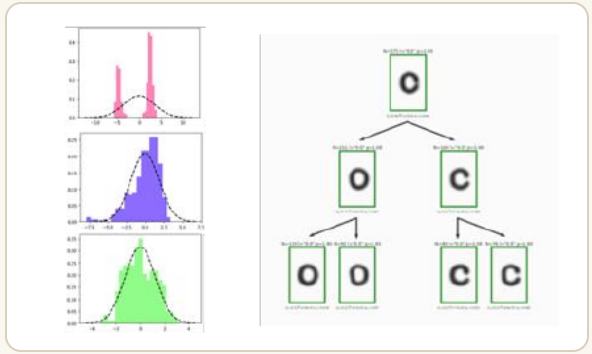
Definition. Extract the shapes of the printed characters from scanned documents, recognise them, and restore them in a printable form. J.-M. Morel, "Reverse typography and the theory of shape: can old books be brought back to life?", FAU MoD lecture, 19 Jun 2026.

- **Why:** a history of typography from the books themselves; old books re-edited with their original appearance.
- **How:** the mathematical theory of planar shapes and its PDEs — where Leon's Mumford–Shah formulation sits.
- Transcription is one stage, not the goal.

Learn the fonts of a corpus, without labels

Idea: printed letters repeat — use that instead of annotating.





One hump: keep. Two humps: split. His slides.

- **Why**: each letter comes from a few metal sorts cast in one mould — one shape per group.
- **Result**: the book’s own font, one prototype per letter — the font stage of reverse typography.

STRENGTH	The redundancy of print is the only supervision, and a cluster can be looked at.
OPEN QUESTION	When is a split a new piece of type, not noise? What about rare letters?

Who printed these plays? The page must say



Title pages of five *sueltas*; colours group the same piece of type across books. D. Belzarena's slides.

40,952 of the **78,524** volumes printed in 17th-c. Spain name no printer, often to evade legal controls. *Sueltas*: plays in quarto, popular, short, on cheap paper.

≈ 5,000 detailed pairwise comparisons an expert would need for about 100 books.

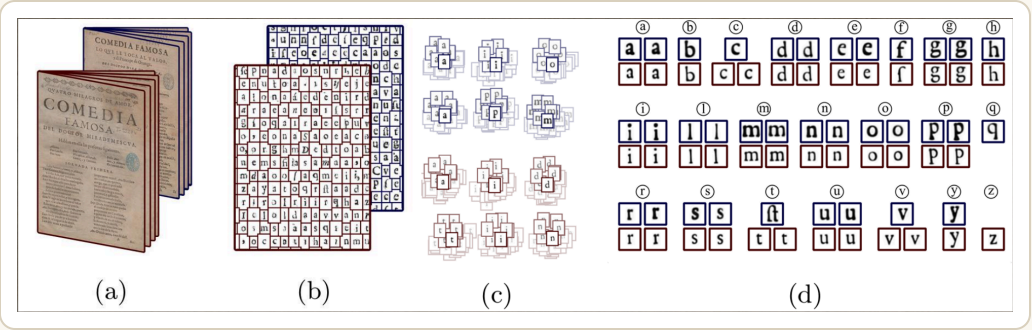
- **Today, by hand**: compare a book's typefaces, ornaments and page layout with books of certain origin; build a network of typographic relations.
- **The trap**: same typeface ≠ same printer — type was traded and shared between shops.
- **Other clues**: watermarks, spacing and spelling habits (Shakespeare's First Folio), damaged sorts.

Reverse typography's question. From the printed page back to the metal type that printed it — and from the type to the print shop.

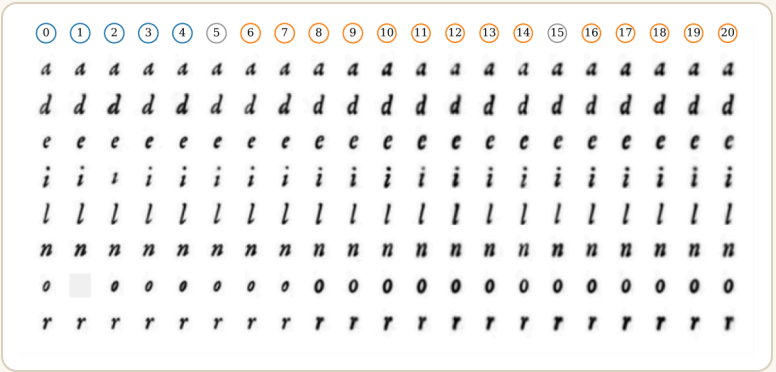
Recover each book's type, then count what differs

Reverse typography in practice: trace each book's letters back to the type that printed them, then compare books type by type.

1. **Find the characters**, roman and italic apart.
2. **One prototype per type, per book**: cluster and average (Mowlavi's method).
3. **Compare two books letter by letter**: distance between aligned prototypes.
4. **Count the letters that surely differ**: more than chance allows if the type were shared (a contrario, 0.01 false alarms).

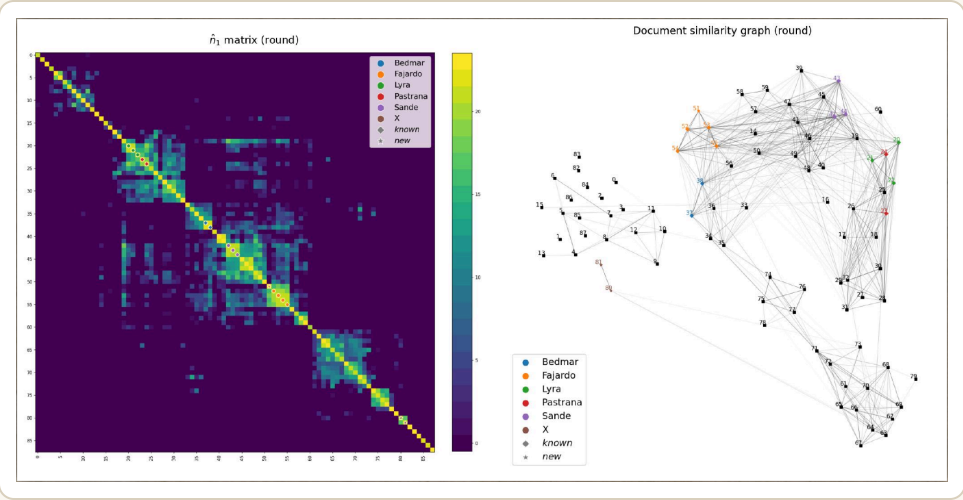


(a) books → (b) characters → (c) clusters → (d) prototypes matched.



Italic prototypes per book: a thin "o" in books 0–7, a bold one in 8–20. Belzarena, Mowlavi et al., 2026.

New attributions, confirmed by experts



88 *sueltas* (BNE): books sharing type are linked; colours mark the 14 with a known printer (Bedmar, Fajardo, Lyra, Pastrana, Sande). His slides.

0.99 AUC for same- vs different-printer on the 91 pairs of known books (roman 0.989, italic 0.997).

book 51 joins Fajardo's family; experts then found shared layout, letterhead type and type damage.

80, 81 a new printing family, perhaps a century apart from the others.

6, 7 moved from Lyra to Rodríguez de Ábrego (their italic, a broken title type); two unattributed books placed.

STRENGTH No training data, a statistical guarantee on every link, and output that experts check by the classical method: about 15 minutes per book instead of thousands of comparisons.

OPEN QUESTION Shared type is not a shared shop: what separates them beyond type (layout, ornaments, damage)? And with no ground truth beyond 14 books, how far does validation scale to the 623 sueltas already collected?

Segmentation has a theory, SAM does not

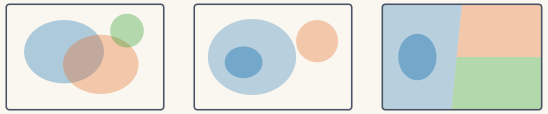
One image: segmentation as an energy

$$E(\mathrm{Seg}) = \sum_{R \in \mathrm{Seg}} \Big(\mathrm{var}(g, R) + \mu \, \mathrm{length}(\partial R) \Big)$$

Mumford–Shah with a constant per region: each region should be uniform, its boundary short. A segmentation is the **minimiser of an explicit functional**, so one can ask whether it exists and how regular its edges are (the 1989 Mumford–Shah conjecture; a proof was claimed in a Sept. 2026 preprint, not yet reviewed).
Raising μ only merges regions: the solutions form a **tree** of segmentations.

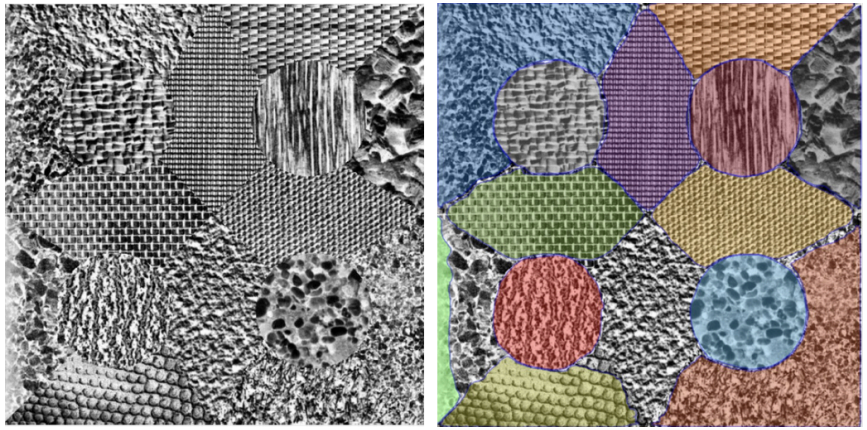
Global models such as SAM: three gaps

1. **Formalization.** A network minimises a loss over its weights, not an energy of the segmentation: existence and regularity cannot even be asked. *Proposal*: an energy on learned features — small variance inside each region, far apart between neighbours — in Mumford–Shah's framework.
2. **Structure.** SAM's masks overlap and leave holes: neither a partition nor a tree. *Proposal*: an overlap energy, zero exactly when masks are nested or disjoint, minimised by merging regions.

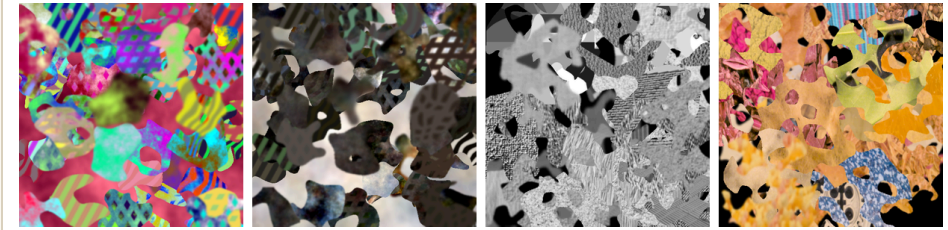


3. **Concept.** "Object" is never defined; it lives in the annotations. Does SAM learn objects, or colour, texture and geometry?

The conceptual gap: test where truth is exact



14 textures, exact boundaries, no semantics. SAM, prompted on a grid, gives **4 of 14** no mask — and nothing in its data says why. His slides.



His generator: random colours · colours picked from a real photo · Brodatz textures · ALOT textures. His slides.

- **Dead leaves**: random shapes stacked on top of one another, sizes following a power law (VibrantLeaves, re-implemented as a modular GPU package).
- **Every factor is a switch**: shapes (hulls, ellipses, polygons), colours, textures, a depth of field with one blur per shape.
- **Exact ground truth, unlimited data**: any error can be traced to a variable.

Abstract SAM: SAM retrained without semantics

For reverse typography, the shape stage: segmentation defined by an explicit energy, learned where the truth is exact — a glyph generator would run the same experiment on print.

	SAM	Abstract SAM	why
image encoder	pretrained on natural images	trained from scratch on abstract images	remove the natural-image prior
prompt	points, boxes	none (stage 1: the true mask)	the signal covers the whole shape
decoder	one object per prompt	the whole image in one pass	no objects to enumerate
loss	focal + Dice on masks	border map + boundary length (+ smoothing)	Mumford–Shah's terms enter training

Status: generator done; stage 1 (encoder from scratch) running, no satisfactory results yet; stage 2 (prompt-free decoder) to do.

STRENGTH Puts global segmentation back into a variational framework where existence and regularity can be asked, and tests SAM where every error is traceable.

OPEN QUESTION Do the proposed energies have minimisers? Will a model trained on abstract leaves segment real images? No results yet to judge.

SYNTHESIS

What recurs, and what stays open

STRENGTHS

- **Reverse typography**: each talk inverts one stage of printing — the font (Mowlavi), the type and the shop (Belzarena), the shape (Leon). Results can be checked.
- **Frugal supervision**: a book's own redundancy, or fully specified synthetic data, replaces annotation.
- **Explanations, not only scores**: clusters one can look at, an a contrario guarantee on every link, an explicit energy.

OPEN QUESTIONS

- **Validation**: only 14 books with a known printer; no results yet for the segmentation model.
- **Transfer**: from 88 to 623 *sueltas*; from leaves to real images, and to glyphs.
- **What a difference means**: another printer, another piece of type, or noise?
- **One inversion**: font, type and scan recovered jointly, to check what a fluent VLM reads.

Reverse typography: when labels are scarce, invert how the page was printed.

THANK YOU

Questions?

BACKUP

Appendix

For questions. Not counted in the seven minutes.

BACKUP

Echoes in my own work

seminar	idea	in my project (historical Chinese prints, TKH)
Mowlavi	cluster characters, correct by agreement	a k-NN graph over every glyph; labels corrected by diffusion on it
Gardella	a contrario NFA	graph edges kept by an a contrario test on glyph distances
Belzarena	Learnable Typewriter	ported, trained on TKH: test CER 1.44 % vs 8.87 % for CHAT; characters seen < 5 times are never read
Leon	specified synthetic data	glyph boxes from CRAFT refined by the SAM 3 tracker; a font-rendered generator would make glyph segmentation a controlled experiment

BACKUP

References cited in the seminars, and the frame

[1] T. Berg-Kirkpatrick et al. *Unsupervised Transcription of Historical Documents.* 2013.

[2] T. Berg-Kirkpatrick et al. *Improved Typesetting Models for Historical OCR.* 2014.

[3] M. Ryskina et al. *Automatic Compositor Attribution in the First Folio of Shakespeare.* 2017.

[4] N. Vogler et al. *Contrastive Attention Networks for Attribution of Early Modern Print.* 2023.

[5] I. Siglidis et al. *The Learnable Typewriter: A Generative Approach to Text Analysis.* 2024.

[6] D. Belzarena, S. Mowlavi, P. Casariego Castiñeira, A. Ulla Lorenzo, G. Randall, J.-M. Morel. *Theatre Chapbooks At Scale: A Statistical Comparative Analysis of Typography.* arXiv:2607.27266, 2026.

[7] J.-M. Morel. *Reverse typography and the theory of shape: can old books be brought back to life?* FAU MoD lecture, 19 Jun 2026.

Also named, without a reference list: SAM, Mumford–Shah, VibrantLeaves, Brodatz and ALOT textures (Leon).